

IPO Series | Astera Labs: Former VP of Global Sales on the Opportunities in AI Networks and the Evolving Competitive Landscape

Primary:

Astera Labs, Inc.

Associated:

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Parade Technologies, Ltd.

Viewpoint:

Former Executive

 Moderator: Kirang Gohil (KG) | Mar 15, 2024 | 32 Min Read

Tim Lauer (TL), Former VP of Sales and Marketing - Astera Labs

★ SUMMARY

Overview

The call, hosted by Kirang Gohil from Guidepoint, features Tim Lauer, a former VP of Sales and Marketing at Astera Labs, discussing the company's strategic positioning in the high-speed connectivity market, particularly in AI clusters. Astera Labs focuses on solving signal integrity challenges associated with high-speed data transfer in computing systems. The discussion covers the company's product roadmap, customer perceptions, and competitive positioning over the next few years.

Key Insights

- **Strategic Positioning:** Astera Labs has established itself as a leader in high-speed connectivity solutions, particularly with its retimers and CXL memory controllers. Tim Lauer emphasizes the company's early recognition of signal integrity challenges and its execution of a comprehensive product roadmap.
- **Customer Perception:** Customers, including major hyperscalers, view Astera Labs positively due to its expertise in signal integrity and its ability to address complex design challenges. The company has built strong relationships with key players in the cloud ecosystem.
- **Competitive Advantage:** Astera Labs' unique features, such as software programmability and diagnostic capabilities, differentiate its products. The company's Switzerland-like positioning allows it to collaborate effectively with various industry players.

Competitive Landscape

Company	Positioning & Strategy	Market Share & Competition
Astera Labs	Leader in retimers and CXL memory controllers; strong relationships with hyperscalers and OEMs.	Commands a significant market share; positioned to maintain 80% market share in key segments.
Broadcom	Announced entry into the retimer market but lacks the established relationships and expertise of Astera Labs.	Faces challenges in gaining market share due to Astera Labs' established position.
Intel	Investor in Astera Labs; collaborates on technology development.	Supports Astera Labs' positioning but does not directly compete in the retimer market.
TI	Exited the retimer market, highlighting the challenges in this space.	No longer a direct competitor.

Applications & Use Cases

Astera Labs' products, including the Aries retimer, Taurus smart cables, and Leo CXL memory controller, are critical in addressing signal integrity issues in high-speed data environments. These products are used extensively in AI servers and cloud infrastructure to ensure reliable data transfer and memory extension capabilities.

Quantitative Insights

Metric	Value	Context
Gross Margin	76%-77%	Among the top quartile for semiconductor companies, highlighting strong product value.
Customer Concentration	Top 3 customers: 70%	Indicates a diversified customer base compared to previous years.

Market Strategy

Astera Labs focuses on maintaining its leadership in high-speed connectivity by leveraging its unique product features and strong industry relationships. The company aims to continue its first-to-market strategy with new generations of PCIe and CXL technologies, ensuring it remains the preferred supplier for major hyperscalers.

Stage & Timing

Astera Labs is positioned in a growth stage, with a strong product roadmap and established market presence. The company's recent S-1 filing indicates plans for further expansion and capitalization on its market leadership in the coming years.

KG: Good morning, everyone. Welcome. My apologies for the delayed start here. Let's get started. My name is Kirang Gohil, and I'm with Guidepoint. Thanks for joining our call on Astera Labs, where we will do a deep dive into the company, its opportunity in AI clusters, its product positioning, and the competitive landscape.

To discuss all this, I'm very pleased to welcome Tim Lauer to the call. Tim is a former VP of sales and marketing at Astera Labs. He was responsible for hiring US and Asia-based sales operations and FAE teams and drove design wins and strategic relationships with hyper scalers, server OEMs and ODMs, CPU and GPU companies.

Prior to Astera, he was with Intel in various sales-related roles. Currently, he is an operating partner at a VC firm. If you have any questions for Tim, please email them to me at Ask@Guidepoint.com and I'll work your questions into the call today. With that, Tim, thank you so much for being with us today, I appreciate it.

TL: Hey, thanks for having me.

KG: Why don't you give us a little more color on your background, and then we'll get right into it?

TL: My background is in electrical engineering. Coming out of university, I went to Intel in California, and there was an opportunity to set up a lab in Taiwan so I moved over to Taiwan. The lab's main objective was working with the Asia ecosystem that's building computers to solve signal integrity issues and thermal issues and this was over 20 years ago.

As each generation of CPU comes out, it's associated with the PCIe bus so generation two, generation three, we're up to generation five. The speed that the CPU communicates with the endpoints doubles. Just by the law of physics, when the speed doubles, it impacts the distance between the CPU and the endpoint, and which you can have a clean signal.

20-25 years ago, we knew that each generation of CPU has it got faster. It was going to introduce some challenges in communicating with endpoints in terms of the physical distance that the copper trace lines that connect these. Anyway, long story short, over my career, I've worked in sales, marketing, and engineering roles.

I left Intel for a period, worked in startups, and came back to Intel when they acquired Infineon for 5G wireless. I was over in Asia again. I ran the sales team for cloud service providers and telecom sales. I worked very closely with the Taiwan and China ecosystems and then I had an opportunity to move back to the US in which I joined Intel Capital, the VC within Intel.

In that role, after we made the investment with Astera Labs, my team and me personally were responsible for working with Astera Labs to onboard them, to help them in any way we could with marketing, sales, and engineering and that's when I met the founding team. After I decided to leave Intel Capital, there was an opportunity to join Astera Labs, and having already worked with them, I was very impressed with the team and where they were headed. That's how I ended up at Astera Labs. Let me stop there to see the questions.

KG: Great Tim, and thanks for that introduction. Before we get into the specifics, I wanted to start at a high level and get your thoughts on the company and their position in the high-speed connectivity market. How do you think they are positioned over the next three to five years?

TL: When I met Astera Labs the first time, there was just talk about the problem they were solving, which is a very important one for Intel regarding this signal integrity challenge as we move on. There was some talk about CXL, but this was several years ago, and after I joined them then they shared the whole roadmap, which they have executed up to this stage.

They knew at an early stage in their company career that there was a whole roadmap of products and problems that they could solve for hyper scaler customers.

That was pretty insightful and they've gone on to execute that and their product strategy, as you can see from the very first product, Aries to the smart cables and then the Taurus with Ethernet even into Leo, which is a CXL memory controller, they can leverage pieces of their products all throughout, especially with CXL. That helped them in those transitions and building credibility as a very small company with the leaders in the cloud ecosystem.

KG: You've dealt with many customers in your time with the company. What is customers' perception of Astera Labs from what you can tell? What are some of the positive or negative feedback that you've received?

TL: First of all, the founding team came from mainly TI. When TI decided to shut down the product line basically for retimers and in that space, signal integrity, these guys were already experts in that area, and they knew that there was going to be a bigger demand just like I talked about, as things get faster on the CPU and things get more dense on boards, there are going to be a need for this.

That group of signal integrity engineers and hardware design folks is a relatively small group of experts in this space. This team knows them all and worked with them very closely. In designing the first product for Aries, they could take into account some things that would speak directly to these engineers and the challenges that they faced.

Early on, like in PCIe generation three, which was for four or five years back, these design engineers for the hyperscalers for the OEMs like Dell, and HP in the server space, nobody wanted to design in a retimer. That's like a band-aid. If you needed that to solve some signal integrity issues, first of all, it costs money. It was complex. It would

cause other issues potentially in latency and that's a last-ditch effort.

As I mentioned, as things got faster and there were more complex and more dense layouts, what would happen is that these very large companies and the signal integrity engineers would give Casey Morrison a call and team and just say, "Hey, we're on a tight product schedule. It looks like we're having some signal integrity issues. Can you help me out?"

That's where Aries and this development board and the built-in protocol analyzer and some of the other features of it really helped out the signal integrity engineers just when they really needed it. That early relationship where they got some feedback on what they needed and then working with them closely as more and more retimers were required as you go to PCIe Gen4 and now with Gen5 has built up a very deep working relationship that speaks to literally all of the major cloud hyperscalers in their ecosystem, which are the ODMs and the OEMs. That's a pretty tight relationship that I would imagine be very sticky going forward.

KG: They filed their S-1 so I wanted to get your thoughts and see if there were any items in the S-1 that stood out to you. Anything that was surprising or interesting that you think is worth highlighting?

TL: I would just say that generally speaking, I've worked with a number of different semiconductor companies and those that are targeting the data center. Generally speaking, if you can get your team set up in Silicon Valley, Washington, Texas, Taipei in Taiwan and then three or four cities in China like Shenzhen, Shanghai, Beijing, and so on and establish the relationships with the top 10 hyper scalers that are mainly in the US and China and their ODMs who build for them, they're mainly in Taiwan and in China and you're going to be able to get 80% of the market if you have the right product.

The other thing is the endpoint guys. We talked about a retimer goes in between a CPU and an endpoint, and the endpoint is basically in four different categories. It's going to be like an AI or GPU processor. It's going to be a memory. It's going to be an Ethernet or some fabric like a switch.

Setting up folks in those locations and having experience working with the right department, which is the signal integrity engineers and the hardware engineers, and also having a product which has a development capability and diagnostics capability, which is like a test tool within the product within Aries puts them in a great position to do what they talk about in the S-1, which basically have wins or have customers within all of the major folks within the cloud ecosystem.

Now, one other thing I'll add to that, that I also see in the S-1 is about their cloud-scale interrupt. Again, the major driver within the PCIe special interest groups that define the next generation of Gen5, Gen6, and then within CXL, the folks that have the most to gain are obviously the CPU, like AMD and Intel, and then the endpoint guys like the Mellanox for networking and Micron for storage and Nvidia and AMD and so on.

The other thing that really gives Astera Labs a Switzerland positioning is the vision and execution of this cloud-scale internal lab. It started with one of the major CPU manufacturers or designers would provide a very early reference design of a prerelease of their newest CPU and Astera Labs would have that to do their own testing and then get permission to be able to test other people's endpoints to help out the CPU vendor.

Once that was established, the other CPU vendor would want to do the same thing so you have this lab which is set up and has all the confidential spaces to keep but then you have a team led by Casey Morrison of Application Engineers that can literally provide free design consulting to these endpoint providers and then also provide feedback to the

CPU vendors.

It's like an unofficial PCIe testing setup that CXL testing will be able to provide for the industry and so that's been a very good way to get all of the newest designs for PCIe Gen5 for the endpoint guys. Astera Labs was one of the first devices out there that supported CXL 1.1. For those of you who don't know, CXL is the Compute Express Link. It's the protocol that runs over the PCIe Gen4 and Gen5 bus and it adds some capabilities for memory extension, memory pooling, and memory caching but Astera Labs had CXL working before anyone on Aries and later on, of course, they have a CXL product for memory extension or memory controller.

Those things are in the S-1 and that's a little bit of a narrative behind how those all come together to, again, put them in a very good position to continue to be the number one supplier of retimers.

KG: One thing that stood out to me was that they generated 76%-77% gross margin over the past couple of quarters. As you know, this level of gross margin is among the top quartile for semiconductor companies. A few years ago, no one would have believed that a retimer company could have 75%-plus gross margin. The question is, what do you think is enabling them to command such high gross margins? Is this sustainable in your view?

TL: I think the thing that's unique, as I mentioned, there are three very differentiating features that they have built-in, and the one we talked about is basically a test tool with the diagnostics capability.

The second is they were the first to make it software programmable with EEPROM so even before the AI chip companies were coming out talking about hardware software co-design, Astera Labs was already architected that into their products. This other piece of the software, which is the fleet management telemetry piece. For servers, every component on the board, you want to be able to have some insight into whether it's early on like we talked about design and diagnostics or it's over time in fleet management. You can start to see and log information.

If you have an API like was designed in Cosmos, it can be all pulled up to a custom suite that the folks at the hyperscalers have to manage their servers long term. Those things make their product unique and speak to what each level of the organization would want to have and that software piece in particular, there are companies out there that sell this software.

I previously worked for a company called LANDESK Software which was separate then was acquired by Intel, then spun out again. That software has value and that's their company.

KG: Is it the software stack that's differentiating and why they can command those gross margins?

TL: It's all of those things combined but the software is free as far as I know and that provides a lot of value. Again, once those APIs are integrated into the top-level software management, which is often custom for the hyperscalers, that's a pretty sticky approach but then just like I mentioned with the cloud interop lab and their Switzerland-like positioning, Astera Labs is in a position to continue to be first to market with PCIe Gen6 and all the CXL capabilities say like Broadcom who made a bold entry into retimers or at least they announced it.

Let's face it, Broadcom has not made a lot of friends with Intel and others so they don't have that same Switzerland-like position nor do they have the expertise and the focus that Astera Labs talks about purpose-built. Like I mentioned, they're talking directly to the people that are going to be using their products so it's the combination of all of those things. It's their products, it's their positioning, it's their people that are experts in focus that will give them an advantage for the foreseeable future.

KG: On customer concentration, in the S1, they said that in 2022, the top three customers were 89% of revenues, with the majority of revenues coming from one guy. In 2023, the top three were 70% and no customer was greater than one-third of sales. Who do you believe are these large customers that Astera Labs works with?

TL: I think the answer to that can be figured out pretty easily, because if you look at the early investors into the Astera Labs, you can see if you know the name of the company gives it away because there were Habana Labs, there's Annapurna Labs and that personal investor into all of those was an investor in Astera Labs, in particular Annapurna Labs.

Those relationships are quite helpful where they landed in different hyperscalers and that would give you a good indication of at least some of them and then they've mentioned also and you can imagine if the top three hyperscalers in the world and they have a nice China team as well. It's universal in terms of the need for PCIe, especially on Gen5 retimers so there's nobody else that can do it that has CXL enabled.

KG: Annapurna Labs was acquired by Amazon. Habana Labs was acquired by Intel.

Now, what is the positioning of Astera Labs at these customers? Meaning are they typically sole sourced for the retimers or do these customers want to work with a number of suppliers? Do you think that position continues to be the case over the next two to three years?

TL: The way it works with hyperscalers is they definitely need to have in almost any large customer but the ones that can command a procurement strategy like large hyperscalers. They definitely have to have a second supplier. When a second supplier is available, they're going to work to get them in but when it comes down to the most important designs getting, which they're going to be the highest volume, that's when the signal integrity engineers, the hardware engineers, the folks that would be responsible for whether a server was successful or not, they're going to be the ones that are going to make the decision.

While there will be a second supplier, for example, say with Broadcom, any new component that's added to a server, there's a certain amount of qualification time. It could be 18 months depending on the importance of it and the readiness of the product. There'll be a long lead time as long as Astera can continue to be first to market, which I think they're well positioned to do as you transition to PCIe Gen6, they're going to be the first supplier, I would say, for a long time even in China because montage their follower, you can tell they're just trying to copy a lot of the same marketing and try to emulate some of the features.

The longevity of their positioning is going to be there and even when a second supplier is introduced, the engineering teams who have worked and been very successful with the Astera Labs are not going to put that retimer that the second source on the high volume servers, they're going to put them on something as a backup in case Astera Labs has some shortage or something like that. That's how that works and that's how it worked with Intel and AMD for a long time until AMD proved themselves in the enterprise space.

They'll continue to have 80 or more percent market share until somebody else can step through and show they have that same focus and relationship and unique features that signal integrity and hardware engineers want.

KG: There's an incoming question from one of the listeners, and I'll just read it out. Large-scale hyperscale clients typically prefer multiple sources. What is a reasonable, steady state share within a customer for Astera as the volume scale up and competitors get closer?

TL: I think I answered that if I understand the question. I think they maintain 80% easily because usually, let's take a large hyperscaler. They're going to have several projects, but for the most important, highest volume servers, they're

going to choose their top suppliers. It's not like they have 20 designs and each one of them has 5%. It's more of like two or three designs or 80% and the other 20%, generally speaking.

KG: In terms of customers, as we look out over the next three, or five years, which customer bucket do you think is going to become the biggest customers for Astera? Is it the hyperscalers? Is it the chip companies, or is it someone like Nvidia who's doing everything? How should we think about the opportunity for Astera at those customers?

TL: Again, just throwing out some numbers that I had looked at that are public, if you say there are roughly from a Gartner like 13 million servers on an annual basis and I believe close to half of those, or maybe, even more, are considered the hyperscalers or cloud service providers, then within the top 10 cloud service providers, you're taking up a large portion of that 13 million.

While working with the hyperscalers directly, and the other large cloud service providers as we talked about how they're set up to do that with people positioned in the key locations, they'll end with working with their ODMs. They're well positioned to continue to be there to win new designs and it's key, though, to have the whole ecosystem where you're talking to the cloud service providers and then you got their ODMs in Taiwan and China and then there's still a HP, Dell, Supermicro, and Lenovo.

They still have a piece of the market for some of the enterprise or smaller cloud service providers that don't spec in their designs with ODMs out of Taiwan and China. All the things are set up for that space and then you start moving into other areas. I talked about how I used to work in the telecom or they call it the comm service provider space. There's a new set of customers or partners in that space like Ericsson's and so on and Huawei that fell under there and then there are different segments of the server market that are utilized in different areas, but that'd be another big large vertical that's slightly different than the cloud space.

Don't forget at least from my understanding and publicly speaking, that electric cars are increasingly becoming a small data center on wheels or a server on wheels just by the size and where endpoints could be located. I would imagine that automotive would require some retimers. I don't know for certain, but just logically speaking, it would seem that would be another growth market but I don't know if I saw that in S-1 or not.

KG: I think they talked about automotive in their retail road show that's available. Maybe spend a minute on the relationship with Nvidia and broadly the use of the retimers in some of the AI servers that are out there.

TL: Well, despite the investment by Intel and all the talk about that, like I mentioned, Nvidia given their products needs a lot of retimers and is a very important partner and close partner. That's Switzerland positioning that I think Astera Labs has done a great job doing establishing with their cloud interop lab.

The trends, especially with generative AI, large language models, first of all, the clustering that's been required or the increase in clustering has been required for training large language models. That topology, that setup that drove some more demand for retimers just you have these large boxes of CPUs and whether you would need some cabling. They transition the Aries retimer into a smart cable which solves some challenges there.

In general, the server market we talked about with general-purpose servers is the largest one but there's also another report in Gartner that shows the trends on the attach rate for AI into the server market. If there were 13 million, what percentage of those servers would be AI servers? That's growing very fast and again, another big indicator on the need for the number of retimers per server.

I know probably the question is going to be out there. It's like, "How do you estimate the number of retimers per these 13 million servers?" I don't know the answer. All I know is that you can do a worst case, base case, and best case, and look at some of the different use cases in the clustering that's going on. Definitely a big indicator is in the number or the percentage of PCIe Gen5 servers because as we mentioned, the speed of the PCIe bus doubled from Gen4 to Gen5 as it does every time of the new generations but that really reduced the distance between the CPU and the endpoint.

With all the density, there's a really high attach rate to Gen5 in terms of retimers and that would give you a good indication of how quickly maybe the ramp on retimers would be. The faster it ramps to Gen5 and the more time it takes for other competitors to break through with their Gen5 designs that work with CXL, the bigger advantage that Astera Labs is going to have for the foreseeable future. Does that make sense?

KG: It does. Is there a way to then think about the content for Astera Labs in these servers or maybe even for an AI server? What typically is the content for Astera Labs?

TL: Well, all three products are available now so you could have a server that has retimers in it that, again, these retimers support CXL. This would be a PCIe Gen5 and then it would also maybe have an AI processor or AI SoCs in there and then it would need to be cabled to another switch and then if it were going to have memory extension through the CXL memory controller, then it would have a Leo. It could have Aries, it could have Taurus cabling, Ethernet cabling and then it could have a Leo product.

I think they show that in some of their diagrams so you technically could have all three products and you could have multiple ones for retimers. I believe you can tell based on the problems that are solved. We talked about retimers or the designers didn't want to use them unless they need to, but it reconditions the signal. It helps for layout on very dense servers and for cabling, if a copper Ethernet cable doesn't do the trick, then you need an extension, then obviously that's Taurus.

The big value that the CXL memory controller Leo provides is that it extends the amount of memory that can be addressed. There's a physical limitation and that's one of the main reasons why CXL was developed. The reason you want to have more addressable memory is directly related often to the use case for large language models. The alternative to extending the memory is adding another server so there's a big value proposition for implementing a CXL memory extension with DDR5.

That chip is more complicated. It's on a different node and I think it's definitely more expensive. You can infer that. The revenue contribution for Leo and I think they say in the S-1, that could be the talk next year. That's a pretty big deal and it solves a different problem and arguably a more valuable problem.

KG: Is there a way to think about an ASP for a retimer versus that for one of the cables that they supply versus the CXL memory controller?

TL: I don't know whether that information is available and I actually don't know the answer.

KG: There's also another incoming question from the client and again, I'll read it out. Astera Labs describes their architecture as being differentiated from traditional IC architectures that are hardware-based. They mention being 50%-60% implemented in software with a parallel processing engine to run multiple software threads. Are competitors using a similar approach or is that really unique to Astera? Is the programmability the reason why Astera wins?

TL: I can speak to things that happened a couple of years ago and then where the others are now. I don't have that much detail in but early on, some of the competitors, Parade Tech was one example. Parade Tech tried to launch a product without this software configurability.

KG: I'm sorry. Just to make sure we don't get confused, this is PCIe retimers?

TL: That's right. I think they were not successful so they had to respin their chip, I believe, and it set them back. This EEPROM configurability aspect of it, I believe, is what they're talking about and Astera Labs is talking about and that has allowed the software programmability of adjusting the lane margining and things like that, I believe.

Yes, it is a definite competitive advantage and I would say that while that is one of the competitive advantages, probably more important is their leadership on CXL. Early on like we mentioned in the retimer, they were able to demonstrate compatibility with all the CXL protocols and they had to do a lot of that design and those things themselves. I believe some of the competitors now are buying some IPs from Rambus and others.

It's not nearly as proven out as Astera Labs being like I said, the cloud interop referee and being able to test it out with endpoints and all of that. I think that's as much as an advantage as their software configurability. I don't know whether Broadcom has that built in but that definitely is an advantage for Astera Labs.

KG: So something to look for is if Broadcom retimers have EEPROM and have that reconfigurability in them.

TL: I just would add one thing, Broadcom maybe made a press release and made the news that they're going to get in this space but don't forget, this retimer and CXL are very critical and they're so critical that Intel needed to see the industry to do this. If Intel could have done it themselves, then they would have but they have a lot of resources from working there and they've tried a lot of things and a lot of adjacencies they talk about, but they couldn't do it and they knew they couldn't do it. TI got out of the business. They're supposed to be experts in this space and they couldn't do it.

ADI and others so the fact that Broadcom, a networking company thinks they can jump in and do this, if it were that easy then other much larger companies would have done this. I look forward to see if Broadcom can actually be successful because it's not really in their wheelhouse. It's outside of what they've proven they can do and they're certainly not as Switzerland to be able to work with the necessary players to be successful long term.

KG: One question on the retimers. In the past, we've seen that market shifts happen when there is a generation shift. It goes from one generation to the next. Right now for PCI, we are on Gen5. Next is Gen6. Talk to us about, first of all, when do you expect Gen6 to be introduced and the lead that Astera Labs has on Gen6. How important is that lead in determining the market share that they can get when Gen6 eventually ramps?

TL: The industry comes together to define the next generations of PCIe as well as within it CXL. Naturally, though, the CPU companies are in the best position to help influence some of the things because their architectures are designed way far out. Again, Intel is an investor. Intel has worked very closely. AMD is also there and now, Nvidia has their own CPU as well and then to some extent Ampere.

I would say Intel probably still going to be in the lead on Gen6, although I don't know but the indication, the ramping of Gen6 will be directly related to the CPU. When the CPU can launch and in order for the CPU to launch the rest of the endpoints, the rest of the industry has to test out their solutions altogether. Again, that's where this cloud SQL interop has been a really interesting position for Astera Labs to help contribute to that and help fix issues and help. Actually, I would imagine some of the future design specifications within the PCIe special interest groups.

KG: Why can't Broadcom catch up or have a meaningful presence when PCIe Gen6 ramps? They're late on Gen5 but how do you think about competition eventually on Gen6?

TL: I'm not saying that Broadcom can't, I'm just saying that I'd heard before that Broadcom are going to. Definitely, it's a big company, and so on and so forth. My point is that it is such an important component in the overall design that if it were that easy, companies like Intel, AMD, Nvidia, and TI would have never gotten out of the business. It takes a special focus and so on.

I think Intel and the rest of the industry just like they want other people to compete with Nvidia, I would imagine there would be some hope that some others can come in and compete but I'm saying that it's not that easy, it's not just a matter of resources and being able to execute in other spaces and networking. It's a special set of skills and relationships and positioning for the company to be successful here.

KG: Maybe switching gears, talking a little bit more about the CXL memory controllers Leo. It would be helpful if we can very briefly lay out why CXL controllers are becoming so important especially in all these AI servers. The opportunity for the memory controllers and where the market is going.

TL: The physical limitation on the amount of memory that can be addressed was one of the main reasons why CXL was developed. When you have large language models, we have large models in general, more memory is going to help the performance of those AI models. One of the things that CXL does is it increases the addressable memory.

In looking at this problem and this is for DDR5 and looking at this problem several years ago, there are a couple of startups that stepped up to say, "Hey, we can do this." I think the reason why Astera Labs was in a really good position to get an audience with the hyperscalers to get inputs on how the requirements for this memory controller was because they'd already demonstrated that they have CXL-compatible retimers.

They already have the logic built in to be compatible with CXL and so we can take that and leverage it into the discussion with we understand PCIe, we understand and can demonstrate that we're compatible with CXL. We want to also build an SoC that is a memory controller for CXL and we'll be able to test it all. We'll be able to work together with our retimers. At that stage, Intel had already invested and was part of the problem solving, getting that audience was key and then obviously it comes down to them being able to execute on delivering this product. I think that's been shown as well that they've been able to do that.

I think I already mentioned that the major use case and the first part of CXL is really memory extension, but there are other use cases related to pooling and then caching. Honestly, I haven't looked into what are driving those, but I think those are a small amount of considering what the memory extension is for large language models.

KG: Why do you believe Astera Labs is well positioned? Why do you think that they'll have the lead in terms of market share? There are other big guys like Marvell and Microchip, even Rambus here. What is so unique that Astera Labs brings and why do you think they'll win?

TL: I think it's just what I talked about. If you're designing a chip for the industry and I've been in other startups, it's not easy to get the top people at hyperscalers who will give you feedback on what you're planning and what they want to see. If you can be usually that's reserved for the larger companies and I think that's where you see Broadcom change their business model at that stage and now you see Nvidia as well saying, "Hey, we'll do engineering services," because what you can learn if you position yourself as doing engineering design services with the hyper scalers as you can learn what they want, what they see and then you can either provide the design engineering services or wait until the hyper scalers maybe aren't able to execute on all that they want to and then take over with your own chip.

That's really the key. If you're going to be long-term in semiconductors, you got to have those relationships with the hyperscalers who will trust you to either build a chip for them or work together building a solution with them.

KG: How much of the business that Astera Labs does with these big hyper scaler customers is customizable parts, meaning parts that are in some way, shape or form customized specifically for hyper scalers versus those that are just general purpose, mass-market solutions?

TL: It depends on what you say customizable, but I don't think that they really build a one-off for a hyper scaler unless it's a large enough volume. They're a small company, relatively speaking. Maybe there's some on the software side that could be customizable, but for chips, you got to have scale for chips. You can't just build a one-off solution. I would say most of their products, unless it's software that's customized are going to be designed with full scale across all types of customers.

KG: How important is pricing here? Broadcom is not known to be aggressive on pricing meaning they're generally considered to be the ones to raise prices and get value but for some reason, if Broadcom decides to be aggressive and offer retimers at a competitive price, then is price that would ultimately determine market share here or are there other factors that are more important than price when the customer decides on a supplier?

TL: Well, if you're talking about retimers, the thing that I witnessed is that when a retimer was required, it was to solve a problem and it was something that was very valued to solve that problem in a short amount of time. Otherwise, there would be more expensive requirements like relaying out that would cost some money but also cost some other time and other validation.

When you need a timer in that sense, it's valued. I think over time, if there's a competitor who can catch up to eventually provide the same value and tools, then obviously that could affect pricing. I don't think that's happened yet from what I can tell so that's on the timer.

On the CXL memory controller, I don't really know at this stage what the relationships are with the memory guys and how they're working together, and how that could impact things. Again, first and foremost, you got to be able to demonstrate it works and it works to the specifications. I think that's the first part that each vendor has to supply.

KG: All right, Tim. We've reached the end of our allotted time here, so let's wrap up the call there. Thank you so much for taking the time and sharing your insights with us. We really appreciate it.

TL: Excellent. Thanks and hope everybody has a great weekend.

KG: You too. Thank you. Bye-bye.

TL: Bye.

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